Chronicle: Technical Specification & Resilient Architecture

Egocentric Episodic Memory Graph with Spatio-Temporal Gating

Project Team | Hackerman X Technical Blueprint

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1 System Overview & Architecture Philosophy

Chronicle is an egocentric, episodic memory engine designed to parse pre-recorded first-person video into a structured, queryable 4D spatio-temporal knowledge graph. Rather than executing computationally prohibitive visual-language foundation models on every frame, the architecture enforces strict **hierarchical motion and geometric gating**. Heavy foundation models (Florence-2, CLIP) are dispatched purely as event-driven workers triggered by deterministic state-machine transitions.

2 The Production Tech Stack

- **Perception & Tracking:** OpenCV (Video I/O, background-masked affine motion estimation via `cv2.estimateAffinePartial2D`), Ultralytics YOLO-World (Open-vocabulary real-time detection), Google MediaPipe Hands (Hand landmarking and contact-point geometry), ByteTrack with CLIP Visual Re-ID reconciliation.
- **Spatial Reasoning & State Engine:** Shapely (Intersection over Smaller Area [IoS], polygon containment, convex hulls), NetworkX (Dynamic spatio-temporal property graphs), SQLite (Persistent indexed storage of topological anchors and episodic edges).
- **Semantic Engine & Foundation Models:** Hugging Face Transformers (`microsoft/Florence-2-base` for zone captioning and composite action verification), OpenCLIP (`ViT-B-32` for visual place recognition and object re-identification).
- **UI, Telemetry & Visualization:** NiceGUI (FastAPI + Vue/Quasar reactive WebSocket event loop), Vis.js (Client-side interactive physics graph rendering), `ui.log` (Deterministic pipeline event stream).

3 End-to-End Ingestion & Processing Workflow

The pipeline processes video sequentially through a non-blocking, decoupled architecture:

1. **Frame Ingestion & Affine Parallax Gating:** Read frame F_t. Mask out dynamic object and hand bounding boxes. Extract static background features and compute rigid transformation $(\Delta x, \Delta y, \theta)$ relative to F_{t-1} using `cv2.estimateAffinePartial2D` with RANSAC. If global motion magnitude $\|\mathbf{T}\| > \tau_{\text{pan}}$, dispatch a Camera Pan Event; if $\|\mathbf{T}\| < \tau_{\text{still}}$ (< 2.5 px), set camera state to `STATIONARY`.
2. **Open-Vocabulary Detection & Landmark Estimation:** YOLO-World detects user-specified targets (*keys, phone, backpack, cup, table, sofa*). Concurrently, MediaPipe Hands estimates 21 egocentric 2D hand landmarks (tracking wrist, palm center, and index fingertip coordinates).

3. **ByteTrack with Appearance-Based Re-ID Reconciliation:** ByteTrack assigns 2D kinematic trajectory IDs. If ByteTrack emits an unassigned detection, the system extracts its 512D CLIP embedding and evaluates cosine similarity against inactive nodes in the Foreground Graph ($S_C > 0.82$). If matched, the track re-binds to the existing entity rather than minting a duplicate node.
4. **Containment & Interaction Evaluation (Shapely):** Calculate the Intersection over Smaller Area (IoS):
$$\text{IoS}(A, B) = \frac{\text{Area}(A \cap B)}{\min(\text{Area}(A), \text{Area}(B))} \tag{1}$$
An interaction is flagged if hand-fingertip coordinates enter the object's convex hull or if $\text{IoS}(\text{Hand}, \text{Object}) > 0.60$ for a temporal threshold $\Delta t > 0.3$ seconds.
5. **Temporal State Machine Progression:** Entity states update based on relative velocity, support surface alignment (y_{max} contact), and camera motion stability.
6. **Composite Temporal VLM Gating:** Upon state transitions entering `INTERACTED` or `CONTAINED`, extract the 30-frame temporal ring buffer. Stitch a 1×3 composite image (Pre-interaction, Contact point, Post-interaction) and query Florence-2 (`<MORE_DETAILED_CAPTION>`) to generate validated action primitives (e.g., `PICK_UP`, `INSERT_INTO`).
7. **Asynchronous Graph Ingestion:** Event packets are tagged with monotonic presentation timestamps (t_{event}) and pushed to SQLite and NetworkX.
8. **Live Telemetry Stream:** The NiceGUI dashboard receives WebSocket broadcasts, driving graph mutations, metric HUD updates, and terminal execution logs without blocking the ingestion thread.

4 The 9-State Resilient Object Lifecycle

To prevent graph explosion and preserve entity continuity, objects transition through a formal deterministic state machine governed by wall-clock presentation timestamps (Δt) rather than frame counts:

State	Formal Operational Definition	Deterministic Transition Trigger
`NEW`	Unanchored visual detection candidate.	Detection emitted with no ByteTrack match and CLIP cosine similarity < 0.82 to existing nodes.
`VISIBLE_STATIC`	Entity stationary on a designated support anchor.	Object velocity $\|\mathbf{v}\| \approx 0$ relative to background for $\Delta t > 0.5$s, support edge verified (y_{max} contact).
`INTERACTED`	Physical manipulation by the user.	Hand landmark containment or $\text{IoS}(\text{Hand}, \text{Obj}) > 0.60$ sustained for $\Delta t \geq 0.3$s.
`VISIBLE_MOVING`	Independent displacement without hand contact.	Relative object velocity $\|\mathbf{v}\| > \tau_{\text{move}}$ while $\text{IoS}(\text{Hand}, \text{Obj}) < 0.10$.
`HELD`	Entity moving in lockstep with egocentric hand.	Track persists, IoS > 0.50, and object centroid displacement matches hand velocity vector.
`CONTAINED`	Entity enclosed within a secondary container.	Object boundary collapses inside container hull (IoS > 0.85); track ceases while container persists.
`OCCLUDED`	Line-of-sight blocked; camera stable.	Track lost; camera motion $\|\mathbf{T}\| < \tau_{\text{still}}$; last known bounding box overlaps occluder hull.
`OUT_OF_VIEW`	Entity exited visual frustum due to head rotation.	Track lost concurrently with camera pan motion ($\|\mathbf{T}\| > \tau_{\text{pan}}$); spatial anchor retained.
`LOST`	Target unobserved across extended duration.	`OUT_OF_VIEW` duration $\Delta t > 10$ minutes. Node flagged dormant; terminal anchor preserved.

5 Dual-Graph Topology & Memory Persistence

- **Background Graph (Topological Anchors):** Nodes represent invariant structural zones (*Kitchen, Work Desk*) and fixed furniture (*Countertop, Dining Table*). When a camera pan settles, a keyframe CLIP embedding is compared against active zones. If cosine similarity > 0.85, the camera relocalizes to the existing zone; otherwise, Florence-2 generates a new semantic zone node.
- **Foreground Graph (Episodic Entities & Relations):** Nodes represent tracked items (*Keys, Phone, Mug*). Edges capture:
  - *Spatial Edges:* `RESTING_ON` (event-gated upon release over support surface), `INSIDE` (containment within convex hull).
  - *Temporal Transitions:* Directed edges logging state evolution with start and end timestamps: $e = (v_i, v_j, t_{\text{start}}, t_{\text{end}})$.
  - *Action Semantics:* Directed multi-edges (`USER --[PICKED_UP @ t=14.2s]--> KEYS`).
- **Data Retention Policy:** Raw pixel buffers are discarded immediately. Micro-movement bounding box trajectories are purged after a 10-minute window. **Crucial Invariant:** The terminal relational anchor (`Entity --[RESTING_ON]--> Anchor`) is permanently retained in SQLite, ensuring that queries regarding long-lost objects resolve correctly.

6 Natural Language Query Retrieval Architecture

Rather than executing brittle graph traversals (such as raw breadth-first search), queries are resolved via a two-stage **GraphRAG** pipeline:

1. **Entity & Anchor Retrieval:** Parse query entities (e.g., *"keys"*) using string matching or CLIP text-to-node similarity. Retrieve the entity's latest active state and its active spatial edge (`RESTING_ON` or `CONTAINED_IN`).
2. **Spatiotemporal Serialization:** Extract the 2-hop relational subgraph surrounding the entity, including its last interaction timestamp, interacting agent, and parent Zone anchor.
3. **Constrained LLM Synthesis:** Format the extracted subgraph into a structured table and prompt a compact language model to generate an explicit natural language answer citeable by timestamp and location.

7 Modular Role Breakdown

Role A: Perception & Kinematics (OpenCV, YOLO-World, MediaPipe, ByteTrack)

- Configure YOLO-World with dynamic target vocabulary and integrate MediaPipe Hands for 2D egocentric landmark extraction.
- Implement background-masked affine motion estimation using `cv2.estimateAffinePartial2D` with RANSAC to distinguish head rotations from static pauses.
- Integrate ByteTrack with OpenCLIP visual embedding re-identification to eliminate duplicate track creation following occlusions.
- Maintain a 30-frame RAM ring buffer (`collections.deque`) emitting structured interaction clips.

Role B: State Machine & Graph Engineering (NetworkX, Shapely, SQLite)

- Construct the 9-state temporal transition engine using Intersection over Smaller Area (IoS) and support surface heuristics.
- Enforce wall-clock timestamp gating (Δt) across all transitions to maintain framerate independence.
- Implement the dual-graph architecture in NetworkX and establish persistent SQLite serialization schemas.
- Build the terminal resting anchor preservation logic and write the 2-hop subgraph retrieval extractor for query resolution.

Role C: Semantic Engine & Multimodal Verification (Florence-2, CLIP)

- Build the OpenCLIP Visual Place Recognition (VPR) module with a 0.85 cosine similarity threshold to deduplicate room zones.
- Implement the 1×3 temporal strip compositor to convert 30-frame interaction buffers into static multi-moment inputs.
- Deploy `Florence-2-base` with `<MORE_DETAILED_CAPTION>` and implement strict JSON output parsing for action classification.
- Build the GraphRAG query generation interface connecting serialized subgraphs to a local language model.

Role D: Reactive Interface & Judge Telemetry (NiceGUI, Vis.js)

- Develop an asynchronous NiceGUI dashboard featuring synchronized video playback and interactive Vis.js graph visualization.
- Implement the real-time `ui.log` telemetry terminal displaying labeled state shifts (`[GEOM]`, `[STATE]`, `[VLM]`, `[GRAPH]`).
- Build an inspectable node drawer showing real-time object image crops, active state histories, and CLIP similarity scores.
- Implement the interactive natural language query console with direct citation of episodic graph timestamps.

8 Core Research Foundations & Theoretical Validation

- **R4DSG (ACM MM 2026):** *Relative 4D Scene Graph Memory for Egocentric Video.* Validates anchor-relative topological mapping without requiring full 3D metric reconstruction. https://arxiv.org/abs/2608.11017
- **VL-KnG (2026):** *Persistent Spatiotemporal Knowledge Graphs for Egocentric Video Reasoning.* Demonstrates sub-second query resolution over long-horizon episodic graphs. https://arxiv.org/abs/2510.01483
- **ESOM (WACV 2026):** *Online Episodic Memory Visual Query Localization with Egocentric Streaming Object Memory.* Establishes streaming heuristics for first-person object tracking. https://openaccess.thecvf.com/content/WACV2026/html/Manigrasso_Online_Episodic_Memory_Visual_Query_Localization_with_Egocentric_Streaming_Object_WACV_2026_paper.html
- **Hand-Object Interaction Foundations (WACV 2024):** *Egocentric Action Recognition by Capturing Hand-Object Contact.* Validates interaction triggering via boundary and landmark containment over standard IoU. https://openaccess.thecvf.com/content/WACV2024/html/Shiota_Egocentric_Action_Recognition_by_Capturing_Hand-Object_Contact_and_object_WACV_2024_paper.html
- **LightMem-Ego:** Validates the three-tier persistence model (Working, Episodic, and Long-Term Archival memory). https://github.com/zjunlp/LightMem-Ego